NOT MEASUREMENT SENSITIVE
MIL-P-85856/22
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS
SILICON-ALUMINUM PLUS POLYMER,
TYPE I, CLASS 1, COMPOSITION W

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Fe	Al	Si	Cu	Others (1)	
				Each	Total
0.8 max.	Bal	11.0 13.0	0.3 max.	0.2 max.	0.6 max.

(1) Percentages by weight are for the metallic component only. Material shall be a blend of 59 - 61 percent by weight of a silicon/aluminum alloy and 39 - 41 percent by weight of completely aromatic polymer system. Polymer shall conform to the following particle size distribution before blending: A minimum of 99 percent by weight shall pass through No. 100 (150 µm) seive; a minimum of 85 percent by weight shall pass through a No. 170 (90 µm) sieve; and a maximum of 25 percent shall pass through a No. 325 (45 µm) sieve.

Particle Size/Distribution

-200 Mesh (-75 µm)	85.0 min.
-325 Mesh (-45 µm)	70.0 to 90.0
-10 µm	10.0 max.

Sprayed Coating Properties

Bond Strength	6.2 MPa (900 psi) min.
Hardness	Rockwell 15Y, 40 to 75.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 25% max.

AMSC N/A FSC 3439
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Deposit Efficiency and Deposition Rate

55% min for A/H_2 @ 3.5 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

Pratt & Whitney Aircraft	PWA 1349
Rolls-Royce Ltd.	MSRR 9507/15
Williams International	WIMS 653
Allison Gas Turbine	EMS 38900A

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/22)

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